

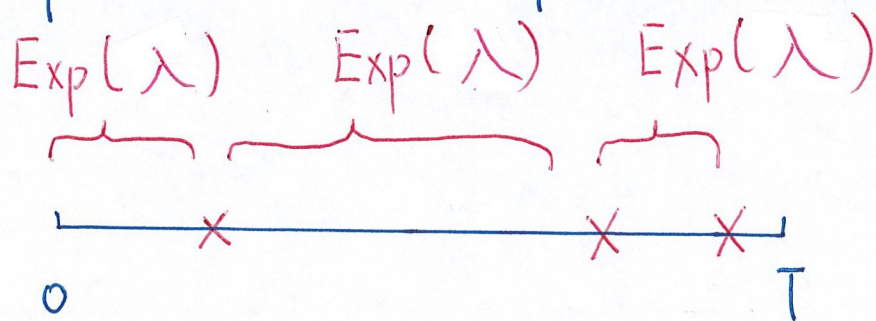
Common continuous distributions

Def. A RV X is said to be distributed exponential if

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & , x \geq 0 \\ 0 & , \text{o.w.} \end{cases}$$

Notation : $X \sim \text{Exp}(\lambda)$, $\lambda > 0$.

Applications : When the # of occurrences of an event is modeled by the Poisson distribution, then the inter-arrival time follows the exponential distribution.



$$\text{ii)} \quad F_X(x) = \int_0^x \theta e^{-\theta x} dx$$

$$= -e^{-\theta x} \Big|_0^x = 1 - e^{-\theta x}$$

$$\text{iii)} \quad S_X(x) = 1 - F_X(x) = e^{-\theta x}$$

$$\text{iv)} \quad M_X(t) = E(e^{xt}) = \int_0^\infty e^{xt} \theta e^{-\theta x} dx$$

$$= \theta \int_0^\infty e^{x(t-\theta)} dx$$

$$= \theta \cdot \frac{1}{t-\theta} \cdot e^{x(t-\theta)} \Big|_0^\infty$$

$\nearrow t-\theta < 0, e^{-\infty} = 0$
 $\searrow t-\theta > 0, e^{\infty} = \infty$

$$= \frac{\theta}{\theta - t}, \quad t < \theta$$

$$\text{ii)} E(X) = \frac{d}{dt} M_X(t) \Big|_{t=0}$$

$$= \theta \cdot (-1) \cdot (\theta - t)^{-2} \cdot (-1) \Big|_{t=0}$$

$$= \frac{1}{\theta}$$

$$\text{iii)} E(X^2) = \frac{d^2}{dt^2} M_X(t) \Big|_{t=0}$$

$$= \theta \cdot (-2) (\theta - t)^{-3} \cdot (-1) \Big|_{t=0}$$

$$= \frac{2}{\theta^2}$$

$$\text{iv)} \text{Var}(X) = \frac{2}{\theta^2} - \left(\frac{1}{\theta}\right)^2 = \frac{1}{\theta^2}$$

▮ Ex. Let X represent the length of a phone call (in minutes), which follows $\text{Exp}(\theta)$. The average length of a phone call is 10 minutes.

$$\begin{aligned} \bullet \quad P(X > 10) &= S_X(10) \\ &= e^{-\theta \cdot 10} \\ &= e^{-1} \end{aligned}$$

$$\begin{aligned} E(X) &= \frac{1}{\theta} = 10 \\ \Rightarrow \theta &= \frac{1}{10} \end{aligned}$$

- Given that a call has lasted for at least 5 minutes, find the probability that the call will last for at least 10 more minutes.

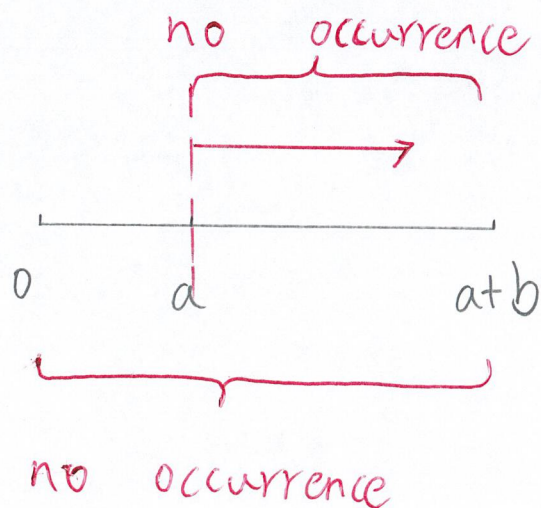
$$P(X > 5 + 10 \mid X > 5)$$

$$= \frac{P(X > 15 \cap X > 5)}{P(X > 5)}$$

$$= \frac{S_X(15)}{S_X(5)} = \frac{e^{-\frac{1}{10} \cdot 15}}{e^{-\frac{1}{10} \cdot 5}} = e^{-1}$$

$$\star P(X > 5 + \boxed{10} \mid X > 5) = P(X > \boxed{10})$$

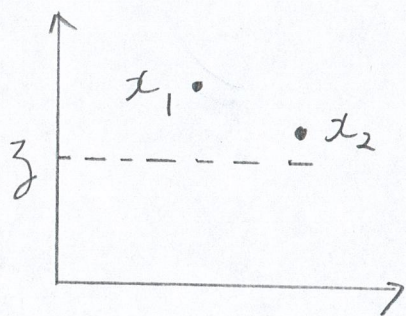
iii) The memoryless property: Assume that $X \sim \text{Exp}(\theta)$,
then $P(X > a+b \mid X > a) = P(X > b)$.



THM: Consider $X_i \sim \text{Exp}(\theta_i)$, which are independent. Let $Z = \min(X_1, X_2)$, then

$$Z \sim \text{Exp}(\theta_1 + \theta_2).$$

$$P(Z > z) = P(\min(X_1, X_2) > z)$$



$$= P(X_1 > z, X_2 > z)$$

$$= P(X_1 > z) \cdot P(X_2 > z)$$

$$= S_{X_1}(z) \cdot S_{X_2}(z)$$

$$= e^{-\theta_1 \cdot z} \cdot e^{-\theta_2 \cdot z}$$

$$= e^{-(\theta_1 + \theta_2) \cdot z}$$

The survival function of $\text{Exp}(\theta_1 + \theta_2)$